Motivation

What this project is about, before any mathematics

Three commitments, each trivial inside its own field, become non-trivial when they meet. Multi-perspective reasoning is the only setting rich enough for complexity to actually develop (modal mathematics); typed data is what makes information robust enough to travel between minds (programming-language theory); unityped agents — participants the system can only see in one way, equal in standing — are how complex collective work avoids collapsing into a hierarchy of experts (concurrency theory). Each is a standard claim in its discipline. The project's conjecture is that these three claims are not independent: they fit into a single mathematical coupling, and the coupling is what gives each of them its force.

The register is the one Alexander Grothendieck named in *Récoltes et Semailles* and Reuben Hersh in *What Is Mathematics, Really?*: not mathematics-as-foundation — the substrate question, what mathematics rests on — but mathematics-as-*praxis*, the practice question, how mathematics is *done*. Praxis, in this reading, is not a finished proof but a community of partial views adjusting one another toward a moving consensus. The project's thesis is that the same mathematical object that captures this community is also the natural setting in which a verifiable communication protocol — between humans, between large language models, between any combination of typed-information-exchanging agents — admits its desirable properties as formal invariants.

That two registers — the philosophical (praxis as community of partial views) and the operational (verifiable protocol for typed agents) — turn out to be the same mathematical object is the surprise the project tries to make precise.

The architectural primitive

Built operationally, the framework is a cascade of trade-offs, each resolved by the same structural move: *separation of concerns*. Locality against globality is resolved by separating the local content layer from the gluing discipline; stability against adaptability by separating the current protocol from its edits; expressive power against transparency by separating how participants communicate from what they communicate about. The cascade is generative: each resolution opens the next problem, and the strategy repeats. The table appears in full inside Section 2; the point here is that separation of concerns is not an engineering principle the framework happens to use — it is the framework's primitive architectural move, applied recursively from the foundations upward.

What this synopsis is

A thesis statement. The two sides of the coupling are sketched in Sections 1 and 2; the coupling itself, with the five names the project hangs on it, is in Section 3; an opening toward field theories — where the same algebraic structure surfaces with a different name — is in Section 4. The companion document *Hoapa: Construction map* shows the formalised pieces in Agda and what they currently support.

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1. The modal side

Modal type theory as a system of perspectives

Modal type theory adds *modalities* — operators that filter, reflect, or restate parts of a structure — to a type-theoretic foundation. The result is a foundation in which different perspectives on the same structure can be made formally precise and held together coherently.

Inspiration

Mid-twentieth-century mathematics underwent a silent revolution, led in different registers by William Lawvere (1937–2023) and Alexander Grothendieck (1928–2014). Mathematics moved from defining objects by their internal construction — *"what is this set?"* — to defining them by their relationships — *"how does this object map to others?"* — and from doing mathematics in one fixed logic to recognising that each context — each *topos* — carries its own internal logic. The shift was from substance to relation, and from one logic to many. Lawvere's particular con-

tribution was *cohesion*: a context-internal distinction between continuous variation and discrete points — the basis of the project's cohesive ∞-topos framework.

Philosophy made a structurally similar move in the same period, finding roots in a lineage traced by Gilles Deleuze: from Duns Scotus and Spinoza, through Leibniz and Nietzsche, to Peirce and Whitehead. This family replaced the search for a single absolute truth with the recognition that reality is irreducibly multi-perspectival, each perspective carrying valid inputs of its own. Both movements arrive at the same embrace of multiplicity. The link between them is structural, not historical: independent disciplines reaching the same place.

Hypothesis

The modal context must be *complex enough* to support reasoning across many levels of structure at once. Yuri Manin (1937–2023), a direct intellectual descendant of Grothendieck, gave a reading of complexity not as randomness, not as regularity, but as *generative depth* — the kind a setting has when it supports multi-level ***adjoint*** reasoning. The hypothesis on the modal side: cohesive modal type theory meets this threshold, and the foundation therefore has enough depth for complexity to actually develop in it; shallower modal structures cannot host this kind of growth.

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2. The concurrency side

Concurrency as implementable protocol

Concurrency studies systems of agents acting in parallel and communicating with each other. Two ideas anchor the modern study of such systems: the ***actor model*** (which describes the agents) and ***session types*** (which describe the structured communication between them).

Inspiration

One of Grothendieck's gifts to mathematics was his radical generalisation of the notion of a *cover* — a collection of local views that together describe a whole, like an atlas mapping a country in patches. Each patch makes sense on its own; the way patches overlap is part of the structure. In his memoirs, *Récoltes et Semailles*, Grothendieck meditates on how mathematical insight grows from local perspectives and the bridges between them; in *Esquisse d'un Programme*, he sketches the structural directions he believed could carry that growth further.

Operational layers

The protocol the project builds separates communication into two layers:

- **"How" layer.** How participants communicate — who talks to whom, in what order, with what kind of messages (question, request, response). The protocol of the conversation itself.
- **"What" layer.** What they communicate about — typed content carrying meaning, structured so every receiver can decode it. The substance of the conversation.

The how-layer is what lets a community of arbitrary participants do something together without imposing a centralised authority on them. The what-layer is what allows their output to be meaningful, readable, reusable and tractable across a network of such protocols.

Hypothesis

The protocol must follow strict *separation of concerns*. The system's organisation is built as the *cascade* the opening section named: each layer's resolution by modular separation opens the next problem; separation of concerns is the recursive move that produces the whole structure.

Layer	Trade-off it navigates	Strategy
Hybrid	locality vs globality	local content glues into global by descent
Reflexive	stability vs adaptability	meta-conversations propose structural edits, consensus ratifies
Bulk	expressive power vs transparency	the protocol generates its own language
Field-theoretic dynamics	exploration vs exploitation	three phases inspired by cosmology: symmetry-breaking, fluctuation, stabilisation
Federation	coherence vs plurality	typed translation between protocols inside a network of protocols
Information cohomology	dense vs accessible encoding	receiver-relative measurement of cost of understanding

Each row's strategy creates the next row's trade-off. The cascade is what builds the whole protocol. The first row — locality against globality, resolved by descent — is the layer the project is currently formalising.

3. The adjunction

How the modal and concurrent sides constitute each other

Call the modal side M (modal contexts: the perspectival structure a foundation carries) and the concurrent side K (concurrent communities: participants, sessions, the act of doing). The relation between them is what category theorists call an *adjunction*: not a single arrow but a tightly coupled bundle of four pieces — two functors and two comparison maps — that cannot be chosen separately. The project hangs five names on the bundle: four for the four pieces, and a fifth for a symmetry that runs across them.

These names are not five independent claims. They are five readings of one structure. Below, each reading carries — in parentheses, in smaller type — the rigorous mathematical object it names; nothing in this section is a metaphor floating free of the math, but the readings themselves are deliberately told in plain language.

The two functors — the gift and the reflection

Complexity
$M \to K$

The modal context, once it is rich enough, unfolds into a community capable of real complexity. This is the *generative* direction — world-creation, the topos as a place where things originate. It is the formal twin of motivation's first claim: a multi-perspective foundation *allows* the kind of typed, unityped collective work the protocol describes; without that permission, the community would be better off collapsing into a hierarchy.

(the *left adjoint* L of the coupling; what the project names *Allow*.)

Evolution
$K \to M$

The community's activity is read back up into the modal structure that makes it intelligible. This is the *reflective* direction — world-understanding, local perspectives gluing into coherent meaning. Iterating it — feed the reflection back into the unfolding and reflect again — produces what the project calls the *developing network*: the long-run sequence of stages a praxis-community moves through. This is where Peirce's "long-run community of inquiry" lands inside the structure, without inheriting its teleology.

(the *right adjoint* R of the coupling; what the project names *Require*. The iteration scaffold is shipped in Agda as the generic developing-network construction; the specific instance waits on the concurrent side's descent layer.)

The two comparison maps — the act and the shadow

Praxis
$id \to R \circ L$

The act by which a modal seed unfolds into doing and is reflected back. It is the *generative comparison map*: not a structure but a movement — the insertion of a perspective into the community where it becomes work, and the return of that work as something the perspective can read. Where Complexity is the gift's possibility, Praxis is the gift in motion.

(the *unit* η of the adjunction.)

Anthropic
$L \circ R \to id$

When one community judges another only through the round-trip "forget what's irreducibly theirs, rebuild what can be rebuilt from outside," the rebuild never recovers the original. The gap is the hidden complexity. Mistaking the gap for zero — believing the forgetful image *is* the whole thing — is the apex-illusion: the structural reason every community can rationally see itself as the summit of complexity while every other community sees the same. The illusion is not a moral failing; it is what happens to any local view that lacks the view from nowhere.

(the *counit* ε of the adjunction, read through its *non-invertibility*; the project's Lawvere-style asymmetry — target T3 — is the formal statement that ε cannot be reversed.)

Two humilities, one fact

The anthropic illusion has a temporal twin. Reading Evolution as *the developing network* means the community is one moment inside an open iteration — not the end of a culmination but a station along an ongoing process. Spatially, the community is not the apex; temporally, it is not the end. These are not two separate humilities. They are the same structural absence — no global frame, no fixed point — cast in two directions. The project's target T3 is precisely the name for that absence: the modal side is rigid enough that its perspectives can be re-applied; the concurrent side is fluid enough that each application produces genuinely new structure. The asymmetry between them is what keeps the iteration alive, and what makes Anthropic an illusion rather than a fact.

Care

The fifth anchor sits orthogonally to the four. It is a symmetry that swaps the generative reading with the reflective one — at the level of typed content, the relation between a protocol and the dual protocol that consumes it: sender of S equals receiver of $S^{\perp}$, the giver-equals-receiver of any typed exchange. Applying it twice returns to the start; it is an order-two move. (in Agda, the proved *dual-involutive* theorem in *Sessions.agda*; the cyclic-operad *boundary-swap* on the interaction side; orthogonal to the four anchors above.)

Outside the mathematics, this involution is the structural form of what developmental psychology and biology have named *attachment* (Bowlby and the attachment literature): the recognition that giving and receiving are the same role under perspective swap — there is no first caregiver, the recursion has no base case, and the asymmetry between roles is local, not foundational. The project does not import attachment theory as content; it notes the structural overlap, because the same order-two swap that proves dual-involutive in our session-types layer is what makes the four-anchor adjunction readable as a home rather than an austere structure.

> *Because nothing sees from the top or the end, you get real power and real humility at once — and care is what makes that a home rather than a void.*

What this shape commits us to, and what it does not

The five anchors are interlocked by the same algebraic discipline that holds any adjunction together: the four pieces cannot be chosen separately, and the order-two involution coherently swaps their two readings. Generations of category theorists have written the coherence laws that pin this discipline; they are what the synopsis points at when it says "five anchors of one structure."

The honest scope: the project has shipped the generic *shape* of the iteration — the developing network as the long-run colimit of any modal unit — in Agda. The *instance* — the specific Complexity / Evolution / Praxis / Anthropic coupling between the cohesive modal side and the actor-causal concurrent side — is the project's central conjecture

and waits on the concurrent side maturing past its current sheaf-descent layer. Mathematically the cardinality of the coupling is also still open: an adjoint *pair* is what the math currently supports; an adjoint *quadruple* (a richer cohesive structure between the two sides) is a possibility the framework is built to extend into without retraction. The synopsis names the five readings; the math will eventually say whether they are anchored on two functors and two transformations, or on four functors with their own coherences.

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4. Toward field theories

An opening, with the cobordism hypothesis as anchor

When the project's typed protocols are read at the right level of abstraction, the same algebraic structure that organises quantum field theories appears. This section does not claim a theorem; it states an opening — the long horizon the framework is built toward — and marks carefully what is grounded, what is targeted, and what is orientation only.

Cup and cap: the geometric face of care

The *cobordism hypothesis* sits inside a particular kind of category — a *compact-closed* one — in which every object has a dual, and the relation between them is encoded by two geometric pieces: a *cup* (a bordism that creates a dual pair out of nothing) and a *cap* (one that annihilates a pair into nothing). It is a categorical theorem that these cup and cap pieces *are exactly* the unit and counit of an adjunction between an object and its dual. Geometrically, orientation reversal — turning a bordism upside down — swaps cup with cap.

That is the geometric face of the order-two involution we called *Care*. In the project's session-typed setting, the involution that takes a protocol to its dual (sender becomes receiver) is the categorical shadow of the same orientation reversal that turns a cobordism around. The "yank the wiggle straight" identity that topologists draw on ribbons is the same coherence as the triangle identity in the adjunction's unit and counit. Care is not metaphorically geometric; it is literally the categorical face of a geometric symmetry.

> *A refinement, not a collapse. The cobordism setting is compact-closed; the project's session-typed layer is finer — a *-autonomous category, in which the two natural ways to combine typed channels (tensor and par) stay distinct. The cobordism reading is the collapsed shadow of session duality, not its full content. We keep the refinement; the cobordism analogy is precise for the duality itself, not for everything around it.*

The modular envelope: where the long-horizon dynamics live

The interaction grammar at the heart of the concurrent side — one binary generator (send/receive), three ports (sender / receiver / channel), a cyclic action that rotates them — is a small example of an algebraic structure called a *cyclic operad*. Its natural closure, obtained by allowing the structure to glue its own marked points to itself, raises the geometry to a stack of stable curves with marked points: $\bar{M}_{g,n}$, the moduli space at the heart of arithmetic geometry. Algebras over the homology of this space have a standard name in mathematical physics: *cohomological field theories*. They are the algebraic skeleton of two-dimensional topological quantum field theory.

This is the long-horizon target on the operational side: the modular closure of the project's cyclic operad, ribbon-decorated to reflect the ordered nature of send/receive, lands on the same geometry the field-theoretic literature has been studying for thirty years. The *field-theoretic dynamics* row of the cascade table in Section 2 is the operational reading of that target: at long times, the protocol's evolution carries the shape of a field-theoretic system on its own combinatorial geometry.

The horizon

Past cohomological field theory, two further horizons are visible from this point. The first is the *cobordism hypothesis* proper (Baez–Dolan; Lurie): the classification of fully-extended topological field theories by fully-dualisable objects in a target category. Read across, it says that the algebraic skeleton of a fully-extended field theory is deter-

mined by a single object with rich duality structure — exactly the kind of object the project's session-types layer is designed to support.

The second is Grothendieck's *Esquisse d'un Programme*: the Teichmüller–Grothendieck tower and the absolute Galois group's action on it, the arithmetic symmetry that lives natively on the moduli space $\bar{M}_{0,3}$ where dessins d'enfants and Belyi maps reside.

> *Orientation, not result. The cohomological-field-theory floor is a precise target the project's modular envelope is constructed to reach; everything past it — the cobordism hypothesis read into the protocol, the Esquisse tower, the Grothendieck–Teichmüller group, and any onward reach toward string-theoretic structures — is the long horizon the framework is oriented by, not a result the project claims. Naming them is how the project marks its direction; not naming them as theorems is how it stays honest.*

Where this leaves the synopsis

The five-anchor coupling of Section 3 is the project's structural centre. The field-theoretic opening of Section 4 is the long horizon it is built toward. Between them lies an enormous amount of mathematics the framework is committed to formalising honestly, layer by layer, in Agda — with the gaps named where they are. The synopsis names what is at stake; the construction map names where the work currently stands; the future is what gets built between the two.

Italic-underlined terms (such as *topos*, *adjoint*, *actor model*, *session types*, *cobordism hypothesis*) are searchable on Wikipedia; in the digital version they are clickable links. The small parenthetical notes in Section 3 give the rigorous mathematical object underneath each plain-language reading. The project's full implementation status, the formalised constructions, and the proposed *Allow* ⊣ *Require* coupling are shown in the companion document *Hoapa: Construction map*.